DATA 590 CAPSTONE FINAL REPORT

FOOTBALL EXPECTED POINTS ADDED: A DIVISION II MODEL

Matthew Klier

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Abstract:

Advanced analytics have taken the sports world by storm over the last few decades or so, and football, while slower to take advantage of the new tools at their disposal, has not been left behind. Fourth down decision making, win percentage probabilities, and new metrics for measuring player production have all been incorporated into the game. Most NFL teams now have their own analytics department and if you go online, you will find no shortage of professionals and amateurs breaking down the numbers. And with so many services like PFF, CollegeFootballData.com, Football Outsiders, and more there is little data to miss for the Pros and big-time college programs.

What most of these services do not have though, is data on the smaller and lower-level programs, like here at Shepherd. While D1 through the pros have access to a near unlimited supply of advanced data, Division II and Division III schools are still in the "Dark Ages." There is no complete database of play-by-play data at either level and there are no advanced statistics models dedicated to these lower levels of play as a result. One of the first and considered most valuable advanced metric is the Expected Points (EP) metric, which actually dates back to the 1970s, which as it sounds is simply a probability model that calculates the expected points to be scored for an offense based on the down, distance, and yard line (Carter & Machol). There are now several different EP models out there for the NFL and Division I Football, and they have been refined much greater since the original. The models are most often used to calculate player or team production in the form of Expected Points Added (EPA) simply the change in EP from play to play, used in place of simply measuring yards gained because it helps to weight the fact that some yards are more valuable than others.

This research sought to take a step forward for the smaller level programs by creating an EP model for Division II football. In order to do so, a Division II play by play database was created for the top 50 Division II teams over the last ten years and a multinomial logistic regression was used to create an EP model specific to Division II. As far as my research has shown, this is the first database of its type and model specific to Division II and hopefully will provide a jumping off point to expand the data collection and analysis for lower level schools.

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EXPERIENCE

SHEPHERD UNIVERSITY

FOOTBALL GRADUATE ASSISTANT *June 2021 - Present*

- Update and automate team financial dashboard
- Create and manage weekly opponent and self-scout dashboards
- Leverage and combine VBA, Power Query, Power Pivot, and base Excel to complete ad-hoc and regular reporting needs

UNIVERSITY OF MINNESOTA

FOOTBALL DEFENSIVE ANALYST *August 2020 – June 2021*

- Manipulated and analyzed data sets with Python, SQL, and Excel
- Created opponent scout breakdowns, statistical reports and dashboards
- Utilized data to analyze and report on relevant metrics and opponent trends
- Automated standard reporting with Power Query
- Developed Python programs to generate visuals of opponent tendencies
- Merged separate data pools to create a central access point for data and reporting

BOTTOMLINE TECHNOLOGIES

NETWORK GROWTH ANALYST *July 2019 – December 2020*

- Created, updated, and distributed automated reports examining business' KPIs
- Leveraged and merged SQL and Salesforce databases to analyze customers and business
- Interpreted and presented visualizations of company data for internal and external usage
- Managed large data sets (over 1 million observations) and developed solutions to ad hoc questions

OLD COLONY YMCA

FINANCE AND ACCOUNTING INTERN *Summer 2018 & 2019*

- Created, analyzed, and managed purchase orders and invoices for accounts payable department
- Communicated with department leaders to ensure timely payments
- Independently developed report analyzing spending patterns of company across the business
- Filed, organized, and maintained company credit card database

COLLEGIATE LEADERSHIP AND ACTIVITIES

COLLEGE OF THE HOLY CROSS

***STUDENT ASSISTANT*, HOLY CROSS FOOTBALL** *September 2017 – May 2019*

- Developed model to track and calculate advanced player statistics from game and practice data
- Created and presented reports on team performance analytics and opponent statistics/tendencies

EDUCATION

College of the Holy Cross

Worcester, MA
B.A. in Economics
May 2019
GPA: 3.87/4.0
Summa Cum Laude
Department Honors Program
Honors Thesis: Athletically Related Aid and Collegiate Athletic Program Success
COES Professional and Probusiness Program

Shepherd University

Shepherdstown, WV
M.S. in Data Analytics and I.S.
Expected: May 2023
GPA: 4.0/4.0
Graduate Assistant Football Coach

SKILLS

- **Microsoft Office (Word, Excel, PowerPoint)**
- **Power Query and Power Pivot, Power BI, Excel Add-Ins, and GSuite Applications**
- **Proficient in SQL, Python, and VBA**
- **Experience with gathering data via Web Scraping and accessing APIs**
- **Proficient in data analysis tools including STATA and XOS**
- **Regression Analysis and other Econometric Statistical Methods**

Introduction:

Ever since the first college football game between Rutgers and Princeton in 1769 intercollegiate athletics and football itself have been a key component of the higher education experience. Athletics play an important role for universities and can help to influence their ability to attract and retain students or bring in alumni donations. Many works, such as Murphy and Trandel (1994), McEvoy (2005) and Pope and Pope (2014), have shown evidence of the "Flutie Effect" where athletic success leads to more undergraduate applications and interest from potential students. Athletic success does not just translate into more applications but an improvement in the quality of applicants. Mixon, Trevino, and Minto (2004), Tucker (2005), and McCormick and Tinsley (1987) all show that success on the football field correlated with higher SAT scores for incoming freshman at universities. Success on the field can also mean an increase for a school's endowment as Booker and Klastorin (1981), Baade and Sundberg (1996), and Rhoads and Gerking (2000) showed with evidence suggesting that successful teams, especially in relation to bowl game and NCAA tournament appearances, were a significant positive factor in determining alumni giving. With all these potential benefits, it is clear why schools would seek to build successful athletic programs.

In order to build successful programs, teams need the right tools to measure what success is. Michael Lewis's bestselling book Moneyball (2004) and its film adaptation from 2011, highlighting the Oakland A's use of mathematical modeling to build and low cost but high preforming roster, was likely the first introduction to advanced sports analytics for many people. Sports Analytics had been around long before Moneyball though. There are many articles dating back to the 1950s about various topics in sports from maximizing yardage on run plays in football, efficiency of right-handed vs left-handed batters in baseball, and an advanced production formula for basketball (Mottley; Lindsey; Heeren). In football though, while analytics have certainly always played a role, in the last decade or so they have certainly exploded in popularity.

Analytics has become a major factor in the modern game of football. They play a role in decision making in game, team building in the offseason, and player development in the weight room. There is no shortage of data now for the NFL and big-time college programs in NCAA Division I. Pro Football Focus provides data, analysis, rankings, and grades for the NFL and NCAA Division I programs (PFF Team Services). Those with access to PFF have a near endless supply of data to work with. nflfastR, is an open source and publicly available package in R that allows anyone with R to access the play-by-play data for NFL teams to perform their own analysis (nflfastR). CollegeFootballData.com provides the same service as nflfastR, bur for NCAA Division I Football (College Football Data). There are many, many more services that provide access to data or analysis surrounding football and if you go online or on twitter, you will find no shortage of blogs and posts from professionals to amateurs performing their own analytical analysis. What you won't find though, is anyone doing so for the smaller levels of football, mainly NCAA Division II or III. Large databases and advanced analysis do not exist at this level of the game, but the structure of the game is no different, and the data does exist for these things to be created. If schools and athletic programs want to get the most out of their athletic programs, then they need to find ways to take advantage and make use of the advanced analytics that the higher levels have been for a long time now.

One of the major and most valued advanced metrics is the Expected Points metric and its use to calculate Expected Points Added. Expected Points (EP) as it sounds is simply an expected value calculated from a probability model that calculates the expected points to be scored for an offense based on the down, distance, and yard line. The first EP model actually dates back to the 1971 when graduate student and NFL player Virgil Carter and his wife estimated the expected points for 1st and ten within different ten-yard splits of the field (Carter and Machol). Bob Carroll, Pete Palmer, and John Thorn created their own linear model in 1988, where every 20 yards changed expected value by 2 points, starting with a team's own goal line being worth -2, their model did not take down and distance into account (Carroll et al.). Since these studies there have been numerous studies and models created. Brian Burke built a model using a locally estimated scatterplot smoothing technique (Burke). ESPN has their own model relying on Bayesian hierarchical statistics (Pattani). PFF, nflfastR, and CollegeFootballData.com all also have their own models and estimates (Gilani). There are certainly more as well, the creation of an EP model for football is not original, many analysts have done so in many ways.

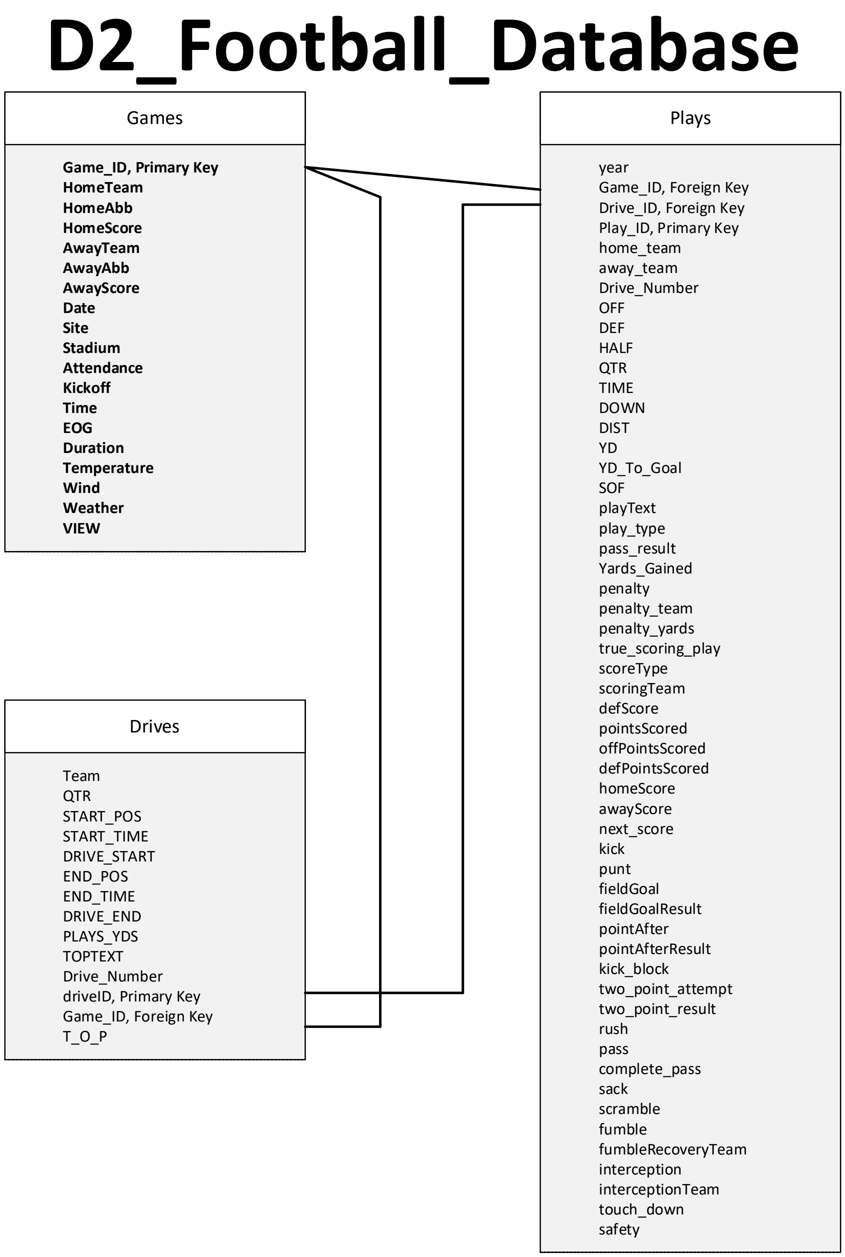
This study differentiates itself from prior research by creating an EP model for NCAA Division II data, using a new database that was created for the purpose of this study. The methodology was not wholly original, and is similar that used by nflfastR creator Saiem Gilani, in the use of a multinomial logistic regression to create a expected value function that determined expected points for a given down, distance and yard line on the field. This study was both able to successfully create a NCAA Division II football play-by-play database with game data available for the last ten years and create an expected points model that can be used to measure team and player efficiency. Both accomplishments still leave room for improvement but are the first of their kind for NCAA Division II and can help serve as a first step for bringing the lower levels of college football into the modern era of analytics.

Data:

The key point of this study is to create a model that can create a probability estimate of how many points a football team is likely to score based on their field position. As such, the key points of data are the field position elements of down, distance to first down or goal line, and yards to the goal line, as well as the scoring results from those field positions. Unlike NCAA Division I and the NFL, there is currently no accessible database that serves as a home for this information at the NCAA Division II level. What does exist is the play-by-play breakdowns for games available on team websites. These play by plays have a basic format providing the down, distance, yard line, and a box of text describing the play. These play-by-play pages then house all the data we need to create the desired model. To collect and convert all of these pages to usable data by hand though would take an incredibly long amount of time, so instead this study relied on a program developed to scrape these webpages.

Using the Beautiful Soup package on Python, a program was written the is capable of parsing the HTML on the play-by-play pages to extract the data in a usable format fast and efficiently. A large majority of college athletic programs, especially those at the NCAA Division II level, use the same software for their websites, meaning that almost all of them have the same HTML format for their play-by-play data. With the HTML format being the same across the majority of college athletic programs, the Python program was able to be created as a one size fits all program. A large amount of data was collected, including many data fields beyond the initial scope of the expected points model, but fields that are certainly relevant to the application of such model. There were also 3 different levels at which data were collected. Game level data was data pertaining to the individual game, such as the date, teams competing, and final score.

Drive level data pertains to one offensive series within a game and holds data such as who the offense and defense were, how and where the drive started and how and where the drive ended. Play level data pertains to the individual play within a drive and houses the data we are most interested in down, distance, and yards to goal line, but also other information such as play type, play result, and gain in yards. All the data was stored in an SQLite database composed of 3 tables for the different levels of data. The schema of the database can be seen below in Picture A.



The game level data held 19 fields, drive data held 15 fields, and play data held 54 fields. While the collection of much of this data is not directly relevant to the creation of the expected points model, much of it would be needed for any good application of the model. Also, with the lack of resources dedicated to data collection for NCAA Division II football, the creation of this database and all the data collected would prove useful in other research and analysis for the sport. As far as this study has found, this database is the first of its kind for Division II football and is just as much an achievement as the model itself is.

The actual data set that was used was a subset of all the data available. This study had a limited time frame to be completed, and although the beautiful soup program allowed for data collection to occur much faster, it would still be an incredible task to collect data for all 169 NCAA Division II football programs. As such, this study specifically targeted 50 teams. The NCAA Division II Football playoffs split the field into four regions and rank and post the top 10 teams in each region. The first 40 teams used for data collection were the teams ranked in these regional top 10 list for the 2022 season. The last ten teams were selected as the next ten top scoring teams from 2022 not already included. The teams selected based off of these qualifications can be seen below in Picture B.

	SUPER REGION ONE	SUPER REGION TWO	SUPER REGION THREE	SUPER REGION FOUR	Next Ten Offenses
1	Shepherd	Benedict	Grand Valley State	Angelo State	CAL PA
2	Assumption	Delta State	Ferris State	Colorado School of Mines	Tiffin University
3	Indiana (PA)	West Florida	Pittsburg State	Minnesota State Mankato	West Chester
4	Ashland	Virginia Union	Ouachita Baptist	Bemidji State	Valdosta State
5	Slippery Rock	Wingate	Davenport	Winona State	Fayetville State
6	Notre Dame (OH)	Mars Hill	Truman State	Wayne State (NE)	University of Charleston
7	Gannon	Fort Valley State	UIndy	Sioux Falls	Central Washington
8	Concord	Limestone	Harding	Colorado State University Pueblo	Albany State
9	New Haven	Tuskegee	Emporia State	Minnesota Duluth	Colorado Mesa
10	Kutztown	West Georgia	Northwest Missouri State	Augustana (SD)	Lenoir Rhyne

These qualifications were chosen to give a list of top competitors in the division but to also ensure that the data encapsulated the sport across the country. As well, although these fifty teams were selected to be the initial points of focus, it does not mean they are the only teams included in the study. Two of these teams, Ferris State and Mars Hill, use different software for their website and as such have different HTML code that the beautiful soup program could not collect the data from their websites. The data for these teams was collected by using their opponents' websites.

For each of these teams, their schedules for the last ten years (where available) were scraped and added to the database. This includes all their games and opponents dating back to the 2013 season, not just opponents off this list of 50. The 2013 season was selected as the cutoff point, as beyond this is where there is a drop off in data availability from team websites, but it still provides this study with nine seasons (2021 there was no real season due to covid) worth of data which is plenty. From scraping the schedules from these 50 teams, 246 unique teams have data collected for them. This number is so much larger than the 169 teams in NCAA Division II as it includes teams from the past who have moved up or down a division and includes teams from other divisions who played against the Division II teams we examined. The data used in the model was only from teams that were active NCAA Division II Football members at the time a game was played.

Although a majority of teams use the same software and share the same HTML format, the data is still entered in by humans and is not in the structure of a database. The format of the data as well as the nature of human error, means that the beautiful soup program can not always correctly or completely collect all the data. Contingencies were written into the program, to skip a game if it came across an error which completely prevented it from collecting the data, these games were then stored in a separate table on the database, so a count and record of games that could not be used was accessible. Over the nine seasons of the fifty teams, 4,230 games were attempted to be scrapped and 228 games were not able to be scrapped, leaving the study with 4,002 games to work off.

The study relies on scoring data as the key dependent variable and as such needs to ensure its accuracy. To ensure that the scoring data is reliable, any games whose game level final scores did not much the play level final scores were removed from consideration, as that would mean there was some error in the play data labeling of scoring. This could happen for numerous reasons, including typos in play text, mislabeling the team who has the ball, or unique situations on the field that are hard for the program to parse through (think double fumble touchdown). There 697 games where the final scores did not match and removing them and leaving us with 3,305 games to be used in the model. Any drives that for some reason or another had more than one unique offense listed were also removed. This again can happen for numerous reasons, but the most often is when the writer of the play by play does not separate the drives. This makes it nearly impossible for the program to decipher the end of the drive and whose ball it is without a human check, meaning the scoring data is likely to be off. 1,608 plays were eliminated due to the multiple unique offenses on a drive issue. Finally, any plays that had missing down, distance, or yard line data (typically data missing one was missing all 3) was removed from the dataset. There are simply cases where this data was for some reason not recorded and there are also play types in football in which it does not exist, primarily kick offs. Any play where this data does not exist is not relevant to the study and any play where this data is missing cannot be used and as such, they are all removed from the data set leaving 247,464 plays. From this group of plays we will also remove what is known as “garbage time” plays or plays where the score is so out of hand that teams are no longer trying to score but attempting to run the clock out. It is standard practice to remove these plays from analysis as the motives of the teams have changed. For this study garbage time will be defined as any play in the 2nd quarter where the score differential is over 38 points, in the 3rd quarter where the score differential is over 28 points, or in the 4th quarter where the score differential is over 22 points. Removing garbage plays leaves 213,497 plays to be used in creating the model.

From the available plays, the dependent variable is the next score after a play and before the end of half or the game. This can be represented numerically or categorically based on the following list: TD (7), FG (3), Safety (2), No Score (0), Opponent Safety (-2), Opponent FG (-3), Opponent TD (-7). The reason we do not simply use how the specific drive a play occurs in ends is because the field position a team leaves another team with may set up the next score. If a team turns the ball over on their own 20-yard line then they likely set up the opponents touchdown or field goal. As well, if a team drives down to the 1-yard line but does not convert, then gets a safety after their opponent takes over, they set that score up. A large part about football is field position, and a large part of the discussion around expected points is that some field positions and some yards are more valuable than others, by using the next score rather than simply the drive end as the dependent variable we take this fully into account.

The independent variables are simply down, distance, and yards to goal line. Down can take a value between 1 and 4, but can be represented as a categorical variable. Yards to goal line is an integer between 0 and 100, where the lower the number the closer an offense is to their opponents endzone. Distance is also an integer between 0 and 100, but it can never be greater than the yards to goal line simply because of the rules of football.

Methodology:

The aim of this study is to create an expected points function that can be used to calculate an expected points value for an offense given their down, distance, and yards to the goal line on a given play in a football game. Although points are integer based, in football, the types of scores are more categorical with integer values attached. The seven potential score types are Touchdown, Field Goal, Safety, No Score, Opponent Safety, Opponent Field Goal, and Opponent Touchdown. As such, the EP function will look like any expected value function, the sum each score type's probability multiplied by its point value, as shown below.

$$7 \text{ X } P_{TD} + 3 \text{ X } P_{FG} + 2 \text{ X } P_{SAF} + 0 \text{ X } P_{NO\ SCORE} + -2 \text{ X } P_{OPP\ SAF} + -3 \text{ X } P_{OPP\ FG} + -7 \text{ X } P_{OPP\ TD} = EP$$

These probabilities will be calculated by using a multinomial logistic regression model, which will run logistic regressions for each score type and then re-weight the probabilities so that they add up to one. The independent variables in the regressions will be the down, distance, and yards to goal line checking for their relationship to predict the probability of the next score. This methodology in itself is not original, and has been used in many different models, and is well documented and explained by Saiem Gilani in his blog. What is original is taking this model and applying it to the new data set that this study has created for NCAA Division II Football.

Methodology 1:

Before diving into the full scale multinomial logistic regressions, the study first sought to highlight the strength of the relationship between our independent and dependent variables. The relationship is intuitive, and you will not likely find anyone arguing that down, distance, and yards to goal line are not correlated to the points an offense will score. But for the sake of highlighting the significance of the relationship, a ordinary least squared regression will be done on the our variables. In this model, the next score variable will be treated as an integer value of "points scored" rather than categorical and to simplify the model for this exercise downs will be treated as an integer value as well. The regression equation can be seen below:

$$\text{POINTS SCORED}_{\iota\tau} = \beta_0 + \beta_1\text{DOWN}_{\iota\tau} + \beta_2\text{DIST}_{\iota\tau} + \beta_3\text{YARD}_{\iota\tau} + \varepsilon_{\iota\tau}$$

ε = Error

The regression was run in python using pandas and the Statsmodels package's OLS function and the results can be seen below in Chart A:

CHART A.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:      next_score_points   R-squared:                       0.115
Model:                            OLS   Adj. R-squared:                  0.115
Method:                 Least Squares   F-statistic:                     9213.
Date:                Sun, 23 Apr 2023   Prob (F-statistic):               0.00
Time:                        11:21:15   Log-Likelihood:            -6.5388e+05
No. Observations:              213497   AIC:                         1.308e+06
Df Residuals:                  213493   BIC:                         1.308e+06
Df Model:                           3
Covariance Type:            nonrobust
================================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
Intercept      7.4596      0.046    161.636      0.000       7.369       7.550
YD_To_Goal    -0.0692      0.000   -151.217      0.000      -0.070      -0.068
DOWN_NUMBER   -0.7676      0.014    -55.433      0.000      -0.795      -0.740
DIST          -0.0901      0.003    -28.681      0.000      -0.096      -0.084
==============================================================================
Omnibus:                    35807.381   Durbin-Watson:                   1.077
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            15332.219
Skew:                          -0.481   Prob(JB):                         0.00
Kurtosis:                       2.107   Cond. No.                         235.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

From the results we can see that all our independent variables have a negative and statistically significant relationship, at the highest level, with points scored by an offense. So higher downs, larger distances to first down, and farther yards away from the goal line all make it less likely that an offense is to score points. Again, this is not something anyone was likely to argue against, but now we can see the results.

We can also visualize these results by looking at our variables graphed against the average points scored for each yard line, down, or distance value. Yard line and down show the clearest relationship in their graphs of Figure A and Figure B, forming obvious negative relationships. For distance, Figure C, we can see that the relationship certainly seems to have a negative trend but gets messy as the distances increase. It is important to remember that these messy points are extremes in the game of football, in our dataset under one percent of plays had a distance over 20 yards. The picture remains similar when we consider distance with its respective down as shown in the Figure D but highlights the negative relationship to points scored for both factors.

FIGURE A.

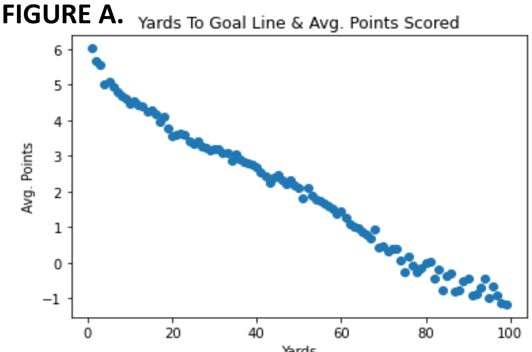


FIGURE B.

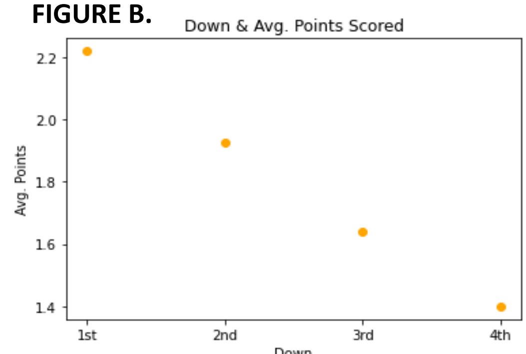


FIGURE C.

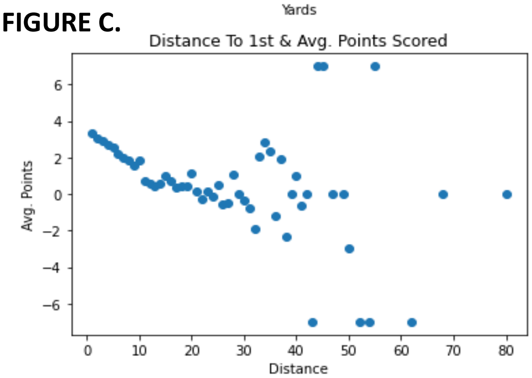
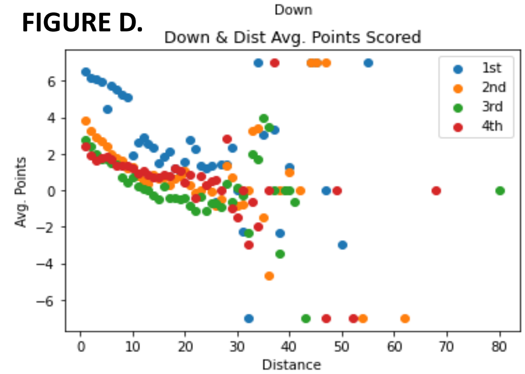


FIGURE D.



While a linear regression can help highlight the strength of the relationships between our variables, it is not the best suited model for estimating points scored. The R-Squared value is only 0.115 and when we examine the results that our regression formula creates, they can fall to an extreme end. We get a maximum expected points of 6.5 using the linear model, which falls within reasonable possibility, but the minimum value falls all the way to -10 points, which we know based on the worst-case scenario being -7 points from an opponent touchdown is not within the realm of possibility. To create a stronger expected points model, we will need to classify our points as categorical binary outcomes and then combine the probability outcomes using a multinomial logistic regression.

Methodology 2:

Using the multinomial logistic regression, six regressions will be run for six of our seven possible outcomes (No Score will be left out as the base case and its probability is the remainder of the sum of all other scoring type probabilities minus one). The regressions are re-weighted to amongst one another to ensure that the probability formulas we get from them ensure that the probabilities will add up to a total of 1, as they are all part of the probability of one event or play. The generic formula for one of the regressions can be seen below.

$$\ln\left(\frac{P(\text{ScoreType}_{\iota\tau})}{P(\text{NoScore}_{\iota\tau})}\right) = \beta_0 + \beta_1 \text{2ND DOWN}_{\iota\tau} + \beta_1 \text{3RD DOWN}_{\iota\tau} + \beta_1 \text{4TH DOWN}_{\iota\tau} + \beta_2 \text{DIST}_{\iota\tau} + \beta_3 \text{YARD}_{\iota\tau} + \varepsilon_{\iota\tau}$$

Downs will no longer be treated as integers but as categorical pieces of data, each with a binary dummy variable representing them, except for first down which will be the reference variable and be calculated when all other dummy variables are set to zero.

Utilizing the multinomial logistic regression method will create a model better suited for creating an EP formula. The creation of probabilities for our scoring categories will ensure that the minimum and maximum EP values are within the realms of possibility. As well, it will account for the fact that the relationship between our independent variables and scoring may not be linear, a fact that we can see in the charts in Figures A and Figure D with some of the bend and curvature in the line. A check was done before running the regression to ensure that there was no serious concern of multicollinearity between our independent variables. All VIF values calculated were below a given threshold of 5, signifying there is no serious concern.

	feature	VIF
0	DOWN_NUMBER	2.448908
1	DIST	3.958576
2	YD_To_Goal	4.264128

	feature	VIF
0	second_down	1.404625
1	third_down	1.155853
2	fourth_down	1.031450
3	DIST	3.791726
4	YD_To_Goal	4.307883

The regression was run on python using the Statsmodels package's MNlogit function and the results can be seen below in Chart B:

CHART B.

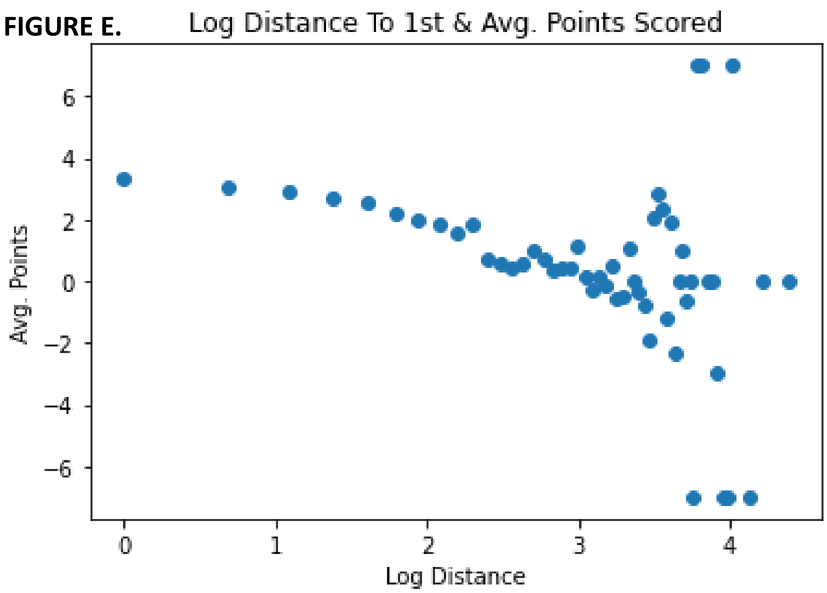
```
                          MNLogit Regression Results
==============================================================================
Dep. Variable:             next_score   No. Observations:               213497
Model:                        MNLogit   Df Residuals:                   213461
Method:                           MLE   Df Model:                           30
Date:                Sun, 23 Apr 2023   Pseudo R-squ.:                 0.06615
Time:                        20:29:10   Log-Likelihood:            -2.8389e+05
converged:                       True   LL-Null:                   -3.0400e+05
Covariance Type:            nonrobust   LLR p-value:                     0.000
=================================================================================
       next_score=FG       coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------
const                    1.5030      0.029     51.957      0.000       1.446       1.560
second_down             -0.0444      0.020     -2.217      0.027      -0.084      -0.005
third_down              -0.3138      0.027    -11.759      0.000      -0.366      -0.262
fourth_down              0.5935      0.036     16.499      0.000       0.523       0.664
DIST                    -0.0042      0.002     -1.937      0.053      -0.008    5.03e-05
YD_To_Goal              -0.0300      0.000    -79.533      0.000      -0.031      -0.029
-----------------------------------------------------------------------------------------
next_score=OPP FG       coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                -1.5460      0.047    -33.033      0.000      -1.638      -1.454
second_down           0.0373      0.027      1.382      0.167      -0.016       0.090
third_down           -0.1861      0.037     -5.050      0.000      -0.258      -0.114
fourth_down           0.2210      0.062      3.548      0.000       0.099       0.343
DIST                 -0.0280      0.003     -8.946      0.000      -0.034      -0.022
YD_To_Goal            0.0125      0.001     22.440      0.000       0.011       0.014
-------------------------------------------------------------------------------------
next_score=OPP Safety       coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------
const                    -7.6949      0.187    -41.069      0.000      -8.062      -7.328
second_down               0.0874      0.083      1.054      0.292      -0.075       0.250
third_down                0.0784      0.110      0.714      0.475      -0.137       0.294
fourth_down               2.3862      0.136     17.589      0.000       2.120       2.652
DIST                      0.0127      0.007      1.746      0.081      -0.002       0.027
YD_To_Goal                0.0571      0.002     25.022      0.000       0.053       0.062
-----------------------------------------------------------------------------------------
next_score=OPP TD       coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                -0.0364      0.029     -1.253      0.210      -0.093       0.021
second_down           0.0432      0.018      2.450      0.014       0.009       0.078
third_down           -0.1318      0.023     -5.666      0.000      -0.177      -0.086
fourth_down           0.2464      0.040      6.178      0.000       0.168       0.325
DIST                 -0.0223      0.002    -11.430      0.000      -0.026      -0.019
YD_To_Goal            0.0112      0.000     31.608      0.000       0.010       0.012
-------------------------------------------------------------------------------------
next_score=Safety       coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                -2.3604      0.122    -19.403      0.000      -2.599      -2.122
second_down          -0.0348      0.082     -0.427      0.669      -0.195       0.125
third_down           -0.3565      0.114     -3.126      0.002      -0.580      -0.133
fourth_down          -0.4972      0.176     -2.826      0.005      -0.842      -0.152
DIST                 -0.0357      0.010     -3.547      0.000      -0.056      -0.016
YD_To_Goal           -0.0147      0.001     -9.897      0.000      -0.018      -0.012
-------------------------------------------------------------------------------------
next_score=TD       coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const             3.2706      0.026    127.103      0.000       3.220       3.321
second_down      -0.3212      0.016    -19.786      0.000      -0.353      -0.289
third_down       -0.8314      0.022    -37.889      0.000      -0.874      -0.788
fourth_down      -1.5180      0.038    -40.425      0.000      -1.592      -1.444
DIST             -0.0777      0.002    -39.988      0.000      -0.082      -0.074
YD_To_Goal       -0.0230      0.000    -75.098      0.000      -0.024      -0.022
=================================================================================
```

From the initial results we do not see many to many surprises. For scoring a TD, all the variables are statistically significant at the highest level and have a negative relationship, so once again, the farther away from the goal line or a first down and the higher the down the less likely a team is to score a touchdown. Field goal we see a similar result, except that fourth down has a positive coefficient, this makes sense, as a team is likely to kick a field goal on fourth down, so a drive which reaches a fourth down would be more likely to result in a field goal. Beyond the results that appear to be obvious with the assumed logic of the game there are a few interesting results. All opponent scores have the strongest relationship for downs with the fourth down indicator, and for opponent safety it is the only statistically significant down factor. Distance has a negative relationship with both Opponent Field Goals and Opponent Touchdowns, which may imply that because teams who have long distances to convert first downs are forced to punt the ball away rather than attempt to go for it, they are able to effectively prevent an opponent score.

The model created with this method provides for a usable expected points formula. It provides for a maximum EP of 5.68 and a minimum of -3.31. Based on most existing models a 5.68 for a 1^{st} down and 1 on the 1-yard line is relatively low, as the touchdown rate from this position at most levels of football is incredibly high, our data set has this field position resulting in the next score being a touchdown 92% of the time. So, there is likely room to improve on the model still. As such, the model was reworked to add in new interaction variables and to try and modify the distance variable to help handle some of the messiness surrounding it as seen in Figure C.

By using the log of the distance, we can hope to limit some of the spread amongst the data and make the relationship to points scored much clearer. This can be seen in Figure E, that shows a much clearer picture of the negative relationship between points scored and distance. While at the tail end for those extreme values we still have some of that messiness, we must again remember these extreme distances are few and far between in the actual game of football.

FIGURE E.



The addition of interaction variables between our down dummy variables and yards to the goal was also added in hopes to improve the model and shed some light on the relationship between how a down is valued based on where it is played on the field. This is especially an point of interest for the fourth down factors weighing so heavily on opponent scores, as we would expect this to be more impactful for down that are played in the teams own territory. It should also be noted that while these are the reasons this study is seeking to implement these adjustments to our model, they are also adjustments and steps taken by previous researchers such as Saiem Gilani in his blog (Gilani) as well as by (another post).

The new regression equation is as follows below:

$$\ln\left(\frac{P(\text{ScoreType}_{\iota\tau})}{P(\text{NoScore}_{\iota\tau})}\right) = \beta_0 + \beta_1\text{2ND DOWN}_{\iota\tau} + \beta_1\text{3RD DOWN}_{\iota\tau} + \beta_1\text{4TH DOWN}_{\iota\tau} + \beta_2\log(\text{DIST})_{\iota\tau} + \beta_3\text{YARD}_{\iota\tau} + \beta_4\text{2ND DOWN}_{\iota\tau} * \text{YARD}_{\iota\tau} + \beta_5\text{3RD DOWN}_{\iota\tau} * \text{YARD}_{\iota\tau} + + \beta_6\text{4TH DOWN}_{\iota\tau} * \text{YARD}_{\iota\tau}\ \varepsilon_{\iota\tau}$$

This regression was also run on python using the Statsmodels package's MNlogit function and the results can be seen below in Chart C:

CHART C.

MNLogit Regression Results

Dep. Variable:	next_score	No. Observations:	213497
Model:	MNLogit	Df Residuals:	213443
Method:	MLE	Df Model:	48
Date:	Sun, 23 Apr 2023	Pseudo R-squ.:	0.06894
Time:	11:47:00	Log-Likelihood:	-2.8304e+05
converged:	True	LL-Null:	-3.0400e+05
Covariance Type:	nonrobust	LLR p-value:	0.000

next_score=FG	coef	std err	z	P>|z|	[0.025	0.975]
const	1.3409	0.043	30.932	0.000	1.256	1.426
second_down	0.0577	0.047	1.225	0.221	-0.035	0.150
third_down	-0.0729	0.059	-1.234	0.217	-0.189	0.043
fourth_down	1.3971	0.069	20.352	0.000	1.263	1.532
Log_Dist	0.0062	0.015	0.414	0.679	-0.023	0.035
YD_To_Goal	-0.0273	0.001	-51.629	0.000	-0.028	-0.026
second_yard	-0.0026	0.001	-3.135	0.002	-0.004	-0.001
third_yard	-0.0059	0.001	-5.305	0.000	-0.008	-0.004
fourth_yard	-0.0356	0.002	-16.307	0.000	-0.040	-0.031

next_score=OPP FG	coef	std err	z	P>|z|	[0.025	0.975]
const	-1.3000	0.070	-18.694	0.000	-1.436	-1.164
second_down	-0.1855	0.082	-2.249	0.024	-0.347	-0.024
third_down	-0.3448	0.106	-3.266	0.001	-0.552	-0.138
fourth_down	0.1421	0.121	1.170	0.242	-0.096	0.380
Log_Dist	-0.1933	0.021	-9.188	0.000	-0.234	-0.152
YD_To_Goal	0.0111	0.001	14.070	0.000	0.010	0.013
second_yard	0.0035	0.001	2.765	0.006	0.001	0.006
third_yard	0.0017	0.002	1.028	0.304	-0.002	0.005
fourth_yard	-0.0014	0.003	-0.481	0.630	-0.007	0.004

next_score=OPP Safety	coef	std err	z	P>|z|	[0.025	0.975]
const	-7.7416	0.316	-24.520	0.000	-8.360	-7.123
second_down	0.4077	0.423	0.964	0.335	-0.421	1.236
third_down	1.5200	0.475	3.202	0.001	0.589	2.451
fourth_down	-0.5834	0.633	-0.921	0.357	-1.825	0.658
Log_Dist	0.1815	0.067	2.690	0.007	0.049	0.314
YD_To_Goal	0.0539	0.004	15.319	0.000	0.047	0.061
second_yard	-0.0036	0.005	-0.655	0.513	-0.014	0.007
third_yard	-0.0189	0.006	-2.969	0.003	-0.031	-0.006
fourth_yard	0.0432	0.008	5.152	0.000	0.027	0.060

next_score=OPP TD	coef	std err	z	P>|z|	[0.025	0.975]
const	0.1586	0.044	3.594	0.000	0.072	0.245
second_down	-0.1233	0.051	-2.418	0.016	-0.223	-0.023
third_down	-0.2951	0.064	-4.619	0.000	-0.420	-0.170
fourth_down	0.2201	0.076	2.891	0.004	0.071	0.369
Log_Dist	-0.1533	0.014	-11.150	0.000	-0.180	-0.126
YD_To_Goal	0.0101	0.001	20.005	0.000	0.009	0.011
second_yard	0.0026	0.001	3.290	0.001	0.001	0.004
third_yard	0.0020	0.001	1.976	0.048	1.61e-05	0.004
fourth_yard	-0.0022	0.002	-1.165	0.244	-0.006	0.002

next_score=Safety	coef	std err	z	P>|z|	[0.025	0.975]
const	-2.1918	0.168	-13.027	0.000	-2.522	-1.862
second_down	-0.0913	0.182	-0.501	0.616	-0.448	0.266
third_down	-0.1977	0.232	-0.853	0.394	-0.652	0.257
fourth_down	0.3505	0.301	1.165	0.244	-0.239	0.940
Log_Dist	-0.2616	0.060	-4.379	0.000	-0.379	-0.145
YD_To_Goal	-0.0133	0.002	-6.411	0.000	-0.017	-0.009
second_yard	0.0005	0.003	0.160	0.873	-0.006	0.007
third_yard	-0.0055	0.004	-1.254	0.210	-0.014	0.003
fourth_yard	-0.0408	0.013	-3.194	0.001	-0.066	-0.016

next_score=TD	coef	std err	z	P>|z|	[0.025	0.975]
const	3.6750	0.037	99.024	0.000	3.602	3.748
second_down	-0.4705	0.041	-11.460	0.000	-0.551	-0.390
third_down	-1.0608	0.052	-20.547	0.000	-1.162	-0.960
fourth_down	-1.7890	0.069	-25.911	0.000	-1.924	-1.654
Log_Dist	-0.5000	0.012	-40.165	0.000	-0.524	-0.476
YD_To_Goal	-0.0239	0.000	-55.791	0.000	-0.025	-0.023
second_yard	0.0024	0.001	3.452	0.001	0.001	0.004
third_yard	0.0020	0.001	2.264	0.024	0.000	0.004
fourth_yard	0.0063	0.002	3.288	0.001	0.003	0.010

The additional regression does not result in any significant changes in the relationships between our independent and dependent variables. In most cases the interaction variables have a very small coefficient not leading to large impacts to the probability. Although, this is not the case for Safeties and Opponent Safeties, where the fourth down and yards to goal interaction variable has a statistically significant and large coefficient, implying that 4th downs deep in a team's own territory are much more likely to result in a safety. We also see fourth and yard to goal interaction take a significant and large negative impact for field goals, helping to balance out the fact that while teams are more likely to kick field goals on fourth down, they will not do so if they are in a field position that puts them out of field goal range. This model also flips the coefficient for distance for field goals to a positive, which also makes sense, with the log helping to limit the extremes, it would make sense that on higher downs with higher distances from converting a first down, teams would opt to kick a field goal.

The adjusted model creates what this study views as an improved expected points model. The maximum EP value is 6.06 and the minimum EP value is -3.13. This puts the model much more in line with other models for other levels of the game. One adjustment needed to be made with the formula. Because Opponent Safeties have large coefficients for third down, fourth down, and distance they get weighed extremely heavily when predicting extreme values, such as a 4th down and 75 having a 70% chance of resulting in safety. While these play situations are realistically of little concern as they are highly unlikely to occur, having such a high probability of an opponent safety, worth -2 points, begins to have the EP values start to increase as the distance continues to get higher, which logically does not make sense. To correct this issue, a cap on the probability of an opponent's safety was set at .25, to ensure that it does not cause the model to break the expected logic. With this adjustment finalized, the EP formula can be implemented for every possible down, distance, and yard to goal combination to create a list of EP values that will be applicable to NCAA Division II play by play data. Numerous methods were used to try and sort this issue with the model out including dummy variables for extreme distances as well as excluding them form the model entirely, but the model still had the same results, and barring further analysis, the safety probability can serve as an effective solution for edge cases that are unlikely to occur in games. Graphs below show the EP values for 1st through 4th and 10 from various yard lines and for 1st through 4th on the 75 yard line with varied distances to give a visual representation of EP.

FIGURE F.

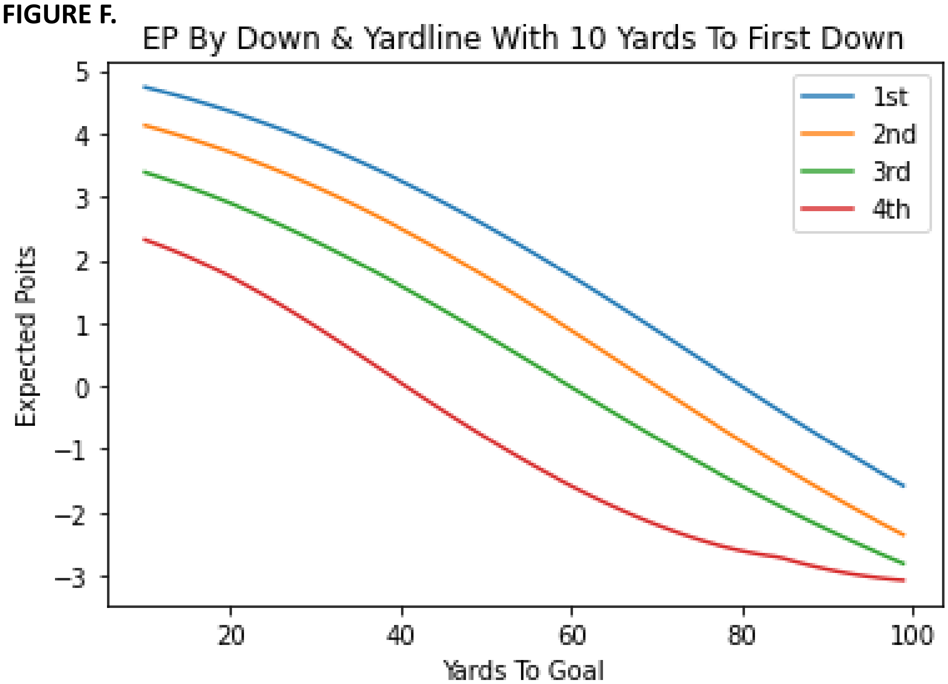
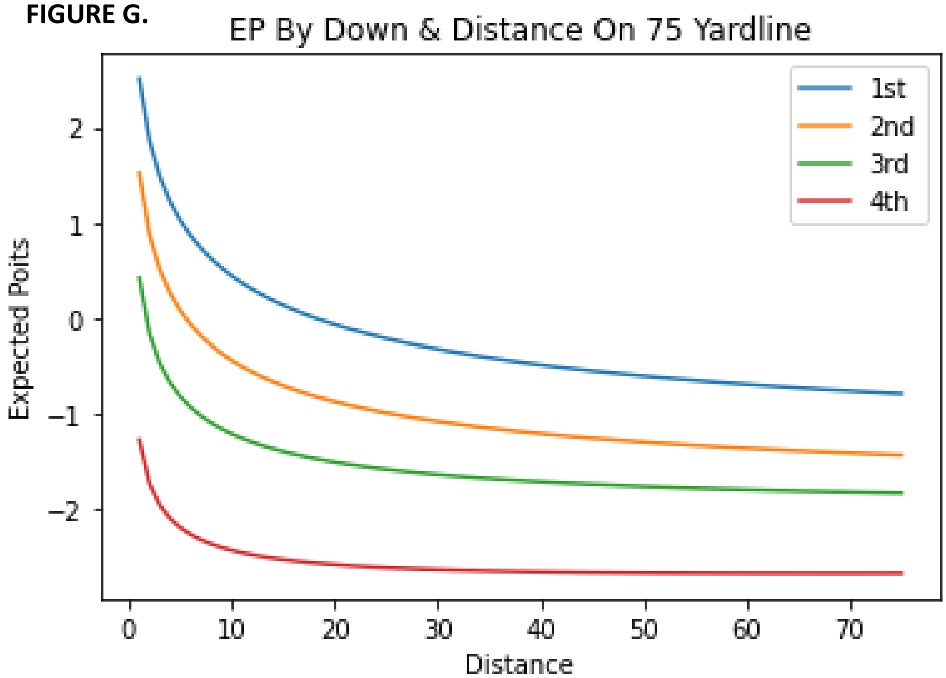


FIGURE G.



Conclusions And Application:

With the world of sports analytics constantly growing at the higher levels of athletics, it is time that the teams at lower levels of the game find a way to take advantage of the data available to them and potentially gain a competitive edge. This study did not create a wholly new original model or new type of metric. Our results are not surprising and do not differentiate from prior research. The basic logic of football still holds true, a team is more likely to score when they have lower downs, lower distances to achieve a new first down, and are closer to the goal line and all these factors play a statistically significant role in their relationship to points scored. What this study did do, is take existing models and modify them to create the first expected data for NCAA Division II Football. As well as creating this model, this study created a new database of play-by-play data for NCAA Division II Football from the past ten years. Both of these advancements should open the door for teams to take advantage of more advanced analytics and for others to pursue new research at the Division II Level.

The EP model can have a quick and easy application as a tool to measure player or team productivity. By simply subtracting the EP from the resulting field position after a play, we can calculate expected points added (EPA) for players. Using EPA is superior to simply measuring yards per attempt or total yards because with the expected model, we have now weighted the value of the yards a player gains. Field position matters a lot in football, as we have seen from this study, and some yards are more valuable than others, so using EPA takes this into account when measuring productivity. We can see a quick application of EPA by looking at the EPAs for the 8 finalists of the 2022 Harlon Hill Award, the NCAA Division II equivalent of the Heisman Trophy.

2022 Harlon Hill Finalists EPA Per Play

From this list of finalists, John Matocha ultimately took home Harlon Hill after leading his team to be the runner up in the NCAA Division II tournament. What we can see though, is that amongst this group of finalists, while Matocha's yards per pass attempt was the most impressive, his EPA per pass attempt was actually in the middle of the pack. This is no dig at Matocha either, he is an incredible player and has numerous other stats and accomplishments from the 2022 season to earn him the Harlon Hill award and there is a lot more that goes into the award than simply one cumulative statistic. This exercise does highlight though how EPA can be used to measure productivity and explosiveness in a different way. Jarod Bowie sticks out right away with the highest EPA and a yards per play stat that is nearly twice as large as the next. What is the most surprising though, is TJ Cole's EPA per rush of 0.57, it is the second highest of the group and while his stats may seem comparable to some of the other running backs in the group, TJ was providing significantly more value to his team.

The application of the EPA model can go far beyond just simply being used as another statistic in determining who the best player in the country is though. For teams, it can be a way to measure efficiency. Teams want their offenses to be efficient, but how do you determine what is an efficient play? Teams have all sorts of different definitions for efficiency. Gain three yards or more on first down, gain half the yards on second, get a first down on third down. Well, a simpler method would be to make sure that the offense is working towards scoring points or, in other words, that the play resulted in a positive EPA. Doing so simplifies the models for what an efficient play is and allows the factors for efficiency to naturally change based on field position and situation. This is just one way that it can be used by teams to help them in analyzing their offensive or defensive success if you reverse the measurement. Teams can use it as another way to measure player productivity, whether based on being a player who directly touches the ball, or by measuring EPA results when a player is on versus off the field. The EPA calculated from the EP model has numerous applications that can help teams and research alike.

Finally, while this work has sought to develop and help bring advanced analytics to NCAA Division II Football, there is certainly room for improvement and further research. While the dataset that was created is the first of its kind for NCAA Division II Football and was able to collect a significant amount of data, there was still a lot of data that was lost or unusable. With more time and further refinement, the beautiful soup program may be able to be improved upon to collect more of this data. Also, some of the data that is unusable due to human error in entering it, may be saved if the work is done by hand. So given the time, we may be able to improve and expand the data set available. There are also many different types of EP models that exist currently, and this study used just one of the existing methods. It is certainly worth seeing how different methods faired in generating different models for Division II. Future research should not just be limited to EP models either, other advanced stats such as win probability, DVOA, yards over expected and many more could also be studied and implemented. The research does not need to end with Division II. Many of the NCAA Division III programs use the same software in creating their play-by-plays that the Division II programs use and as such, the same experiment could be conducted for Division III, further helping to expand analytics to all levels of football.

Professional Growth:

This project was a great opportunity for my own personal growth, but it did not come without its challenges. The biggest challenges were in the creation of the program to scrape the data from the NCAA Division II Football teams' websites. There is a significant amount of research and articles available about expected points models in football and how to create them. There is not a significant amount of information on how to scrape data from 50 different college football teams. While almost all the teams do use the same HTML format, there were still so many asynchronies between how some teams entered the data that a lot of adjustments needed to be made to ensure the program was flexible enough to collect a usable amount of data. Luckily with enough trial, error, and a lot of stack exchange a workable program was finished that collected a lot of data. Enough to work well beyond the realms of this project, which is extremely exciting as there is a lot of potential in what can be done with it. The refinement of the regression model and especially the handling of the extreme cases also proved a significant challenge. Given more time, I would love to dive deeper into why the model ends up putting such a high weight on opponent safeties in extreme cases, but none of the attempts made during this study could prevent the result. Ultimately, it became clear that given the fact that these strange results were only occurring in situations that are so unlikely to occur on a football field, that simply changing the model with a cap to normalize it would work for this scenario.

The project required me to test and acquire a significant amount of technical skill. While I had worked with or used every program or software that I used in this project at some other point, either in class work or employment, I had never used any of them to this depth. From creating the beautiful soup program, I feel much more confident in my Python knowledge and my HTML knowledge. I also feel I have far and improved my SQL skills, having now created a database that had data input into it from Python and then read it back out into Python. As well, I certainly feel better in my ability to search and clean a dataset for bad data, as a long portion of this project was checking to make sure that the data being used was accurate and complete. Finally, I am more confident in my statistical analysis abilities having now implemented a multinomial logistic regression that I was able to successfully turn into an expected value function.

Beyond the technical skills the project helped me in further developing many soft skills. Time management was a big one, as there was only one final deadline for the project and no set class or work time. It was important to set my own goals and deadlines and to make sure I was consistent in meeting them throughout the semester to stay on track. Problem solving and critical thinking are also a given with a project of this type. There were many roadblocks where I simply needed to step away, take a break, and then come back and think of a new solution which may be outside of the realm I was initially thinking.

I would not have been able to complete this project if it weren't for the strong base I have built from my classes at Shepherd. Statistical Analysis, Mathematical Modeling and Big Data

Analytics all helped me prepare for the mathematical portion of developing and implementing a regression model. They also ensured that I was familiar with many of the coding techniques I needed to use to complete the project. Web Programming helped ensure that I had the working knowledge of HTML to be able to scrape the data set from the web. It also provided me with experience using Beautiful Soup as part of my final project in the class. My work in Database Management Systems also came in use in allowing me to create a SQLite database to hold, clean, and access the data being used for the project. All of the work I have done at Shepherd helped me in completing this project and has helped to prepare me for graduation and moving on into the workforce after this semester.

Recap:

This study sought to build upon the large existing pool of works on expected points models for football and open the doors to analytics for the lower levels of the game by creating a play-by-play database and expected points model for NCAA Division II Football. In these respects, the study was a success. A database was successfully created and using a multinomial logistic regression model and EP model was created and applied to NCAA Division II Football data. No shocking results were discovered in the data, down, distance, and yards to the goal line are all still statistically significant in predicting points scored even at the Division II level. Teams want to have lower downs, shorter distances to first down, and be closer to the opponent's goal line to have a higher likelihood of scoring points. The creation of this database and metric for Division II though opens up the possibility for further research and analytical development for Division II and Division III football, to help get all levels of football on a more equal footing when it comes to analytics.

Bibliography

Baade, Robert A., and Jeffrey O. Sundberg. "What determines alumni generosity?." Economics of Education Review 15.1 (1996): 75-81.

Brooker, George W., and Theodore D. Klastorin. "College athletics and alumni giving." Social Science Quarterly 62.4 (1981): 744.

Burke, Brian. Expected points. Advanced Football Analytics (formerly Advanced NFL Stats). Retrieved April 24, 2023, from http://www.advancedfootballanalytics.com/2008/08/expected-points.html

Carroll, Bob, Pete Palmer, and John Thorn. "The hidden game of football." The Hidden Game of Football. University of Chicago Press, 2023.

Carter, Virgil, and Robert E. Machol. "Operations research on football." Operations Research 19.2 (1971): 541-544.

College Football Data. CollegeFootballData.com. (n.d.). Retrieved April 24, 2023, from https://collegefootballdata.com/about

Gilani, S. (2020, June 16). Series on sports analytics: College football expected points model fundamentals. Tomahawk Nation. Retrieved April 24, 2023, from https://www.tomahawknation.com/florida-state-football-fsu-noles/2020/6/16/21212025/series-on-sports-analytics-college-football-expected-points-model-fundamentals-part-3

Heeren, Dave. The basketball abstract. Prentice Hall, 1988.

Lewis, Michael. Moneyball: The art of winning an unfair game. WW Norton & Company, 2004.

Lindsey, George R. "An investigation of strategies in baseball." Operations Research 11.4 (1963): 477-501.

McCormick, Robert E., and Maurice Tinsley. "Athletics versus academics? Evidence from SAT scores." Journal of Political Economy 95.5 (1987): 1103-1116.

McEvoy, Chad. "The relationship between dramatic changes in team performance and undergraduate admissions applications." *The SMART Journal* 2.1 (2005): 17-24.

Mixon, Franklin G., Len J. Treviño, and Taisa C. Minto. "Touchdowns and test scores: Exploring the relationship between athletics and academics." *Applied Economics Letters* 11.7 (2004): 421-424.

Mottley, Charles M. "The application of operations-research methods to athletic games." Journal of the Operations Research Society of America 2.3 (1954): 335-338.

Murphy, Robert G., and Gregory A. Trandel. "The relation between a university's football record and the size of its applicant pool." *Economics of Education Review* 13.3 (1994): 265-270.

nflfastR. nflfastr.com. (n.d.). Retrieved April 24, 2023, from https://www.nflfastr.com/

Pattani, A. (2012, September 15). Expected points and EPA explained. ESPN. Retrieved April 24, 2023, from https://www.espn.com/nfl/story/_/id/8379024/nfl-explaining-expected-points-metric

PFF Team Services. PFF. (n.d.). Retrieved April 24, 2023, from https://www.pff.com/team-services

Pope, Devin G., and Jaren C. Pope. "Understanding college application decisions: Why college sports success matters." *Journal of sports Economics* 15.2 (2014): 107-131.

Rhoads, Thomas A., and Shelby Gerking. "Educational contributions, academic quality, and athletic success." Contemporary Economic Policy 18.2 (2000): 248-258.

Tucker, Irvin B. "Big-time pigskin success: Is there an advertising effect?." Journal of Sports Economics 6.2 (2005): 222-229.